The coupled field equations of the 7+1 model and their solutions in the three gravity regimes: provenance

Sources, data, decisions, derivations and the verification record of the action term by term, the coupled field equations, the radion, and the accurate approximate solutions in the primordial, modern and far-future regimes — sections 20–23 of the notebook 7+1.nb

Repository: github.com/once-ere/7plus1 — prepared for Nash
Notebook: 7+1.nb (Wolfram Language 15.0.1)
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Abstract

This record states what was asked, which data were used and where they come from, every decision taken with the equation behind it, how each result was derived and how it was checked: the 48 checks of sections 20–23 of 7+1.nb (48 True), the 49 boolean checks of the independent Python program tools/ref_coupled.py (49 True), the comparison of the 34 groups of numbers printed by both programs, the program versions and the SHA-256 hashes of the files.

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1. The task

On 27 October 2026 the author asked for the following (summarized; the original wording is the author's private working material and is not reproduced):

1. Compute accurate approximate solutions of the coupled field equations of the model for all three gravity regimes.
2. Write each and every interaction in equation form, define all symbols, and explain and analyze the solutions — in a documentation file with a companion provenance file and a symbol file, each in Markdown and LaTeX, the LaTeX compiled to PDF.
3. Afterwards analyze, in documents of the same kind, whether dirac16complex can be the dark matter (and the dark energy), on the basis of these solutions (a separate stage with its own documents).
4. Never refer the reader to another file; put every equation that is created or used into the file that uses it.
5. Push every completed stage to github.com/once-ere/7plus1 (respecting the repository's .gitignore) and verify the repository; continue through all stages without stopping.

Interpretation of "the three gravity regimes". The model has three epochs in which gravity behaves in qualitatively different ways, the same three epochs that section 14 of 7+1.nb introduced: (I) the PRIMORDIAL regime near $x_0 = 0$, where the higher Lovelock terms dominate — inside it the cubic, the Gauss–Bonnet and the Einstein term take over one after another, so regime I is solved in three sub-regimes; (II) the MODERN regime, where the auxiliary space is held by its stabilizing potential and gravity on the brane is four-dimensional; (III) the FAR FUTURE, dominated by the four-dimensional cosmological constant. This reading is stated at the top of the document and of section 20 of the notebook.

The coordinates of 7+1.nb are kept: $X = (x_0, \dots, x_7)$, x_0 the time (the direction of the octonion unit $e_0 = 1$), x_1, x_2, x_3 ordinary space, $x_4, \dots, x_7$ the four auxiliary space-like coordinates.

1.1 The signature: the author's explanation of 27 October 2026

Source. On 27 October 2026 the author of the project explained that the xAct list $\{7, 1, 0\}$ declares seven space-like dimensions and one time-like dimension, while $\mathrm{diag}(+1, -1, \dots, -1)$ is simply the squares of the octonion basis, and asked that this be made clear in all documentation, examples and provenance records. **What it corrected.** The version of 7+1.nb pushed in the commits 52d8729 to dc2f178 of github.com/once-ere/7plus1 declared the xAct metric of section 12 with `DefMetric[{1, 7, 0}, …]`, reading the list as the signs of η; since then section 12 declares `DefMetric[{7, 1, 0}, g71[-a, -b], CD, …]`, exactly as the author's notebook Einstein-Lovelock-7+1-Nash.nb does. **What was added.** Checks in 7+1.nb — section 4.7: η is the squares of the octonion basis, $e_A e_A = \eta_{AA} e_0$ and $\mathrm{Re}(e_A e_B) = \eta_{AB}$; section 12.2: `SignatureOfMetric[g71]` $= \{7, 1, 0\}$ with `SignDetOfMetric[g71]` $= -1$, $\mathrm{Re}(e_A e_B) = \eta_{AB}$, and $\mathrm{Re}(x\,x) = \eta_{AB} x^A x^B$; section 17: the flat metric $\mathrm{diag}(+1, -1, -1, -1)$ of the world brane is the squares of the basis e_0, e_1, e_2, e_3 of the quaternion subalgebra — and the same statements in the Python programs `tools/ref71.py`, `tools/ref_gravity.py` and `tools/ref_sm.py`; the explanation itself in the text of 7+1.nb (sections 1.2, 1.4, 4.2, 4.7, 7, 11, 12.2, 13, 15.1, 17) and in every document, symbol file and provenance file of the repository. **What did not change.** No result. Compared output by output with the backup below, every output of 7+1.nb is the same, except the declared list itself, the new checks and the check counts (and, as in every run, the name of the private context and the measured times); in particular every component of every Lovelock tensor is unchanged. The components follow from the component matrix of the metric, and the sign of the determinant, `SignDetOfMetric`, is -1 for both lists. **Backup.** The evaluated notebook as it was before is kept in the workspace as `backups/7+1_2026-10-27_before_signature.nb` (the folder backups/ is excluded from the repository by .gitignore), 1034401 bytes, SHA-256 `060942488caf4f45120ffd37be0123e96992a31cd7fdbde09ee4dc4ced4f5d3a`; `tools/verify_7p1_nb.py` compares every input, text and section cell of 7+1.nb with it and finds every cell present, each one unchanged or changed by this explanation, and it compares every output in the same way.

The explanation, in full. The 7+1 spacetime has one time-like dimension (x_0, the direction of the octonion unit $e_0 = 1$) and seven space-like dimensions (x_1, x_2, x_3 ordinary space, $x_4, \dots, x_7$ auxiliary space). Two different objects describe this, and neither may be replaced by the other. (i) The tensor package xAct, used in section 12 of 7+1.nb exactly as in the author's notebook Einstein-Lovelock-7+1-Nash.nb, declares the metric with `DefMetric[{7, 1, 0}, g71[-a, -b], CD, …]`: the list $\{7, 1, 0\}$ declares seven space-like dimensions, one time-like dimension and no degenerate dimension — the numbers of entries $+1$ and -1 of $\eta_{71} = \mathrm{diag}(-1, +1, \dots, +1)$ (`η71`), the same spacetime written in the sign convention of 4+4.nb (space-like $= +1$, time-like $= -1$). (ii) The matrix $\eta = -\eta_{71} = \mathrm{diag}(+1, -1, \dots, -1)$ (`η17`) is simply the squares of the octonion basis: $e_0^2 = +1$ for the unit $e_0 = 1$ (time) and $e_i^2 = -1$ for the seven imaginary units $e_1, \dots, e_7$ (space); more generally it is the real part of the product of two basis units, and it gives the quadratic form of the octonions:

$$\eta_{AB} = \mathrm{Re}(e_A e_B), \qquad e_0^2 = +1, \qquad e_i^2 = -1 \quad (i = 1, \dots, 7),$$
$$\mathrm{Re}(x\,x) = x_0^2 - x_1^2 - \cdots - x_7^2 = \eta_{AB}\, x^A x^B \qquad (x = x^A e_A \in \mathbb{O}).$$

η gives the signs of the Clifford relation $\Gamma_A \Gamma_B + \Gamma_B \Gamma_A = 2\,\eta_{AB}\, 1_{16}$ of the real 16×16 Dirac matrices: time squares to $+1_{16}$ and each of the seven space directions to -1_{16}, exactly like the octonion units (in the notation $Cl(p, q)$, with p generators squaring to $+1$ and q to -1, the algebra $Cl(1, 7) \cong \mathbb{R}(16)$). It also gives the signs of the metric components: $g_{00} > 0$ along the time-like direction, negative squares along the space-like directions. Every component computed by 7+1.nb follows from the component matrix of the metric; the declared list fixes for xAct the sign of the determinant of the metric, `SignDetOfMetric[g71]` $= -1$, which is the same in both sign conventions, since $\det \eta = \det \eta_{71} = -1$. 7+1.nb checks `SignatureOfMetric[g71]` $= \{7, 1, 0\}$, `SignDetOfMetric[g71]` $= -1$, $\mathrm{Re}(e_A e_B) = \eta_{AB}$ and $\mathrm{Re}(x\,x) = \eta_{AB} x^A x^B$.

2. Inputs

2.1 Objects of the earlier sections of 7+1.nb

Sections 20–23 use, unchanged: the coordinates `X`, the metric `η17` $= \eta$, the check registry `check[name, test]`; the curvature engine `curvature[g]` (section 11), the pair-set Lovelock densities `lovelockL` and tensors `lovelockE` with `rudAssoc` (section 12.5); the closed forms `ECos` of the Lovelock tensors of the cosmological metric and their sums `eTot[c]` $= \sum_k \alpha_k E^{(k)c}{}_c - \Lambda$ (section 12.6; $c = 1, 2, 5$ are the 00, 11, 44 components), with `toH` replacing the derivatives of a and b by H_a, H_b, H'_a, H'_b; the Kasner forms `EKas` (section 14.1). The closed forms are, with $H'_a = dH_a/dx_0$:

$$
\begin{aligned}
E^{(1)0}{}_0 &= 3\left(H_a^2 + 4H_aH_b + 2H_b^2\right), \\
E^{(1)1}{}_1 &= 2H'_a + 4H'_b + 3H_a^2 + 8H_aH_b + 10H_b^2, \\
E^{(1)4}{}_4 &= 3H'_a + 3H'_b + 6H_a^2 + 9H_aH_b + 6H_b^2,
\end{aligned}
$$

$$
\begin{aligned}
E^{(2)0}{}_0 &= -12H_b\left(4H_a^3 + 18H_a^2H_b + 12H_aH_b^2 + H_b^3\right), \\
E^{(2)1}{}_1 &= -32H'_aH_aH_b - 48H'_aH_b^2 - 16H'_bH_a^2 - 96H'_bH_aH_b \\
&\quad - 48H'_bH_b^2 - 32H_a^3H_b - 136H_a^2H_b^2 - 192H_aH_b^3 \\
&\quad - 60H_b^4, \\
E^{(2)4}{}_4 &= -12H'_aH_a^2 - 72H'_aH_aH_b - 36H'_aH_b^2 - 36H'_bH_a^2 \\
&\quad - 72H'_bH_aH_b - 12H'_bH_b^2 - 12H_a^4 - 108H_a^3H_b \\
&\quad - 180H_a^2H_b^2 - 108H_aH_b^3 - 12H_b^4,
\end{aligned}
$$

$$
\begin{aligned}
E^{(3)0}{}_0 &= 360H_a^2H_b^3\left(4H_a + 3H_b\right), \\
E^{(3)1}{}_1 &= 72H_b^2\left(8H'_aH_aH_b + 2H'_aH_b^2 + 12H'_bH_a^2 + 8H'_bH_aH_b + 8H_a^3H_b + 19H_a^2H_b^2 + 8H_aH_b^3\right), \\
E^{(3)4}{}_4 &= 72H_aH_b\left(9H'_aH_aH_b + 6H'_aH_b^2 + 6H'_bH_a^2 + 9H'_bH_aH_b + 9H_a^3H_b + 17H_a^2H_b^2 + 9H_aH_b^3\right).
\end{aligned}
$$

Before sections 20–23 were appended, the evaluated notebook (then sections 0–20, section 20 being the summary of all checks) was copied to `backups/7+1_2026-10-27_before_coupled.nb` (excluded from the repository by .gitignore); tools/verify_7p1_nb.py checks that every input and text cell and every output of sections 0–19 is unchanged.

2.2 Data and their sources

quantity	value used	source
$M_{\rm Pl}$ (reduced)	2.435×10^{18} GeV	standard value, $(8\pi G_N)^{-1/2}$
M_8	10 TeV $= 10^4$ GeV	choice: the lowest eight-dimensional Planck scale compatible with collider limits on large extra dimensions
α_2, α_3	$1/M_8^2$, $10^{-6}/M_8^4$ ($\xi = 10^{-6}$)	choice: Lovelock couplings of natural size; ξ small separates the sub-regimes
m_r	10 GeV (regime II); $10^{-3}/\ell_2$ (capture run); $0.1/\ell_2$ (de Sitter test)	choices; the formulas hold for any m_r
f_W	M_8^4	choice (natural brane tension)
m_b, m_c, m_t; m_u, m_d, m_s	4.183, 1.2730, 172.56 GeV; 2.16, 4.70, 93.5 MeV	PDG 2025 (as in section 17 of 7+1.nb)
m_e, m_μ, m_τ	0.51099895 MeV, 105.6583755 MeV, 1.77693 GeV	PDG 2024
m_W, m_Z, m_H	80.3692, 91.1880, 125.20 GeV	PDG 2025
hadrons $\pi^\pm$, π^0, $K^\pm$, K^0, η, ρ, ω, K^*, η', p, n	139.57039, 134.9768, 493.677, 497.611, 547.862, 775.26, 782.66, 895.5, 957.78, 938.27209, 939.56542 MeV	PDG 2024
α_s at 10 GeV	0.18	$\alpha_s(M_Z) = 0.1180$ run to 10 GeV (PDG 2024), rounded

quantity	value used	source
QCD crossover	$T_c = 0.160$ GeV, width 0.030 GeV	HotQCD: $T_c = 156.5 \pm 1.5$ MeV; the tanh switch is a model
$N_{\rm eff}$	3.044	standard-model value (Bennett et al. 2021)
$T_{\rm CMB}$	2.7255 K	Fixsen 2009
H_0, Ω_m	67.4 km/s/Mpc, 0.315	Planck 2018 (TT,TE,EE+lowE+lensing); age 13.787 ± 0.020 Gyr for comparison
Ω_b	≈ 0.049 ($\Omega_b h^2 = 0.0224$)	Planck 2018 (only quoted in the table of all interactions)
$\Omega_c h^2$	0.120 (so $\Omega_c = 0.120/0.674^2 = 0.264$)	Planck 2018 ($\Omega_c h^2 = 0.120 \pm 0.001$)
$\hbar$; $\hbar c$; k_B	$6.582119569 \times 10^{-25}$ GeV s; $1.973269804 \times 10^{-16}$ GeV m; $8.617333262 \times 10^{-14}$ GeV/K	CODATA 2018 (exact SI values for $\hbar$, k_B)
1 Mpc; 1 Gyr	$3.0856775814913673 \times 10^{19}$ km; 3.15576×10^{16} s	IAU 2015 (pc); Julian year
g_* for $T_{\rm RH}$	10.75	γ, $e^{\pm}$, three neutrino species

Full references: S. Navas et al. (Particle Data Group), Phys. Rev. D 110 (2024) 030001 and the 2025 update; Planck Collaboration, Astron. Astrophys. 641 (2020) A6; D. J. Fixsen, Astrophys. J. 707 (2009) 916; J. J. Bennett et al., JCAP 04 (2021) 073; A. Bazavov et al. (HotQCD), Phys. Lett. B 795 (2019) 15; N. Arkani-Hamed, S. Dimopoulos, G. Dvali, Phys. Lett. B 429 (1998) 263 and Phys. Rev. D 59 (1999) 086004.

Software: Wolfram Language 15.0.1 (front end WolframNB.exe; the kernel configuration "Eval71" is a fresh kernel); Python 3.13.17 with numpy 2.5.3, scipy 1.18.1, sympy 1.14.0, mpmath 1.3.0.

3. Decisions

#	decision	reason / equation
1	the three regimes are the primordial, modern and far-future epochs; regime I is split into its cubic, Gauss–Bonnet and Einstein sub-regimes	the epochs of section 14 of 7+1.nb; the Lovelock hierarchy $\alpha_3 H^4 \gg \alpha_2 H^2 \gg 1$, $\alpha_2 H^2 \gg 1 \gg \alpha_3 H^4/\alpha_2 H^2$, $\alpha_2 H^2 \ll 1$
2	the homogeneous metric $ds^2 = dx_0^2 - a^2 d\vec{x}^2 - b^2 d\vec{y}^2$ for all three regimes	the symmetry of the model: ordinary space and auxiliary space each flat and isotropic
3	every source written through its comoving energy $W(a,b)$	$\varepsilon = W/(a^3 b^4)$, $p_a = -\partial_a W/(3a^2 b^4)$, $p_b = -\partial_b W/(4a^3 b^3)$ is then conserved automatically
4	units $\ell_2 = \sqrt{\alpha_2} = 1$ and $\xi = \alpha_3/\alpha_2^2 = 10^{-6}$ in regime I	the vacuum equations then contain only ξ; crossover times scale as $\tau_{32} = \lvert q_2\rvert\sqrt{30\xi}\,\ell_2$, $\tau_{21} \approx \ell_2$
5	$\Lambda = 0$ in regime I	$\Lambda \ell_2^2 \approx \Lambda_4/M_8^2 \approx 10^{-92}$
6	the vacuum solution as an explicit curve $X(u)$ plus one quadrature	the constraint is quadratic in $X = H_a^2$ at fixed $u = H_b/H_a$
7	numerical integration in $\sigma = -\ln H_a$ with x_0 as a dependent variable	H_a decreases monotonically; the variable spans 22 decades evenly
8	dirac16complex in regime I as a massless collisionless gas with an isotropic comoving momentum distribution, $C_g = 3 \times 10^{-6}$	the massless, weak-coupling limit of the Kohn–Sham matter of section 16; $\kappa\varepsilon = 10^{-6} \times 3H_a^2$ at $a = b = 1$, $H_a = 1$
9	the size of a curvature term = sum of the absolute values of its monomials	in a Kasner regime the monomials of the dominant term cancel; the net value would hide it
10	stabilizing potential $U = U_0 + \frac{K_U}{2}\ln^2(b/b_0)$	the simplest potential with a minimum at b_0; $m_r^2 = \kappa K_U/12$
11	capture run: $m_r = 10^{-3}/\ell_2$, start at $H_a = 10^{-2}$, b_0 where the Kasner contraction would arrive at $\tau = 1/m_r$	the radion starts displaced by $O(1)$, the generic case
12	τ_E from the average of H_E over one period, then A, ϑ by linear least squares	a three-parameter nonlinear fit is ill-conditioned in τ_E
13	physical units $M_8 = 10$ TeV, $m_r = 10$ GeV, $f_W = M_8^4$ for regimes II and III	natural values; every formula is given for general parameters
14	Standard Model plasma as ideal gases with exact Bose/Fermi statistics, a tanh switch at the QCD crossover, neutrinos decoupled while $e^{\pm}$ are relativistic	accurate to about 10% near the crossover and much better elsewhere
15	$t(T)$ by quadrature on 481 points in $\ln T$, $1/H_a$ exponential in $\ln a$ on each step	exact where g_*, g_{*s} are constant; halving the grid is the error estimate
16	composite late-time solution $t_{rm} + t_{m\Lambda} - t_m$	two exact two-component solutions minus their common matter part

#	decision	reason / equation
17	radion in de Sitter by linearizing the full Einstein–Lovelock equations, δH_a removed with the constraint	gives friction and mass with all Lovelock corrections
18	the independent Python program uses its own closed forms (from tools/lovelock71.py), its own quadratures and an 8th-order Runge–Kutta method (DOP853)	no code is shared with the notebook

4. Derivations

The cubic Lovelock density in invariant form. Expanding

$$L_3 = \frac{1}{8}\delta^{\alpha_1\beta_1\alpha_2\beta_2\alpha_3\beta_3}_{\gamma_1\delta_1\gamma_2\delta_2\gamma_3\delta_3} R^{\gamma_1\delta_1}{}_{\alpha_1\beta_1} R^{\gamma_2\delta_2}{}_{\alpha_2\beta_2} R^{\gamma_3\delta_3}{}_{\alpha_3\beta_3}$$

over the 720 permutations gives eight invariants with the coefficients

$$\begin{aligned} L_3 = {} & R^3 - 12RR_{\mu\nu}R^{\mu\nu} + 16R^{\mu}{}_{\nu}R^{\nu}{}_{\rho}R^{\rho}{}_{\mu} + 24R^{\mu\rho}R^{\nu\sigma}R_{\mu\nu\rho\sigma} + 3RR_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma} \\ & - 24R^{\mu\nu}R_{\mu}{}^{\rho\sigma\tau}R_{\nu\rho\sigma\tau} + 4R^{\mu\nu}{}_{\rho\sigma}R^{\rho\sigma}{}_{\lambda\tau}R^{\lambda\tau}{}_{\mu\nu} - 8R_{\mu}{}^{\rho}{}_{\nu}{}^{\sigma}R_{\rho}{}^{\lambda}{}_{\sigma}{}^{\tau}R_{\lambda}{}^{\mu}{}_{\tau}{}^{\nu}, \end{aligned}$$

an algebraic identity, checked exactly on four random integer tensors $R_{abcd} = \sum_{n=1}^{3}(S^{(n)}_{ac}S^{(n)}_{bd} - S^{(n)}_{ad}S^{(n)}_{bc})$ ($S^{(n)}$ symmetric with entries in $[-6,6]$), which have all the algebraic symmetries of a Riemann tensor; the same tensors check $L_2 = R^2 - 4R_{\mu\nu}R^{\mu\nu} + R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma}$.

Energy–momentum from $W(a,b)$. For the homogeneous metric with lapse n, a source with comoving energy $W(a,b)$ has the action $S_m = -\int nW\,dx_0$ (per unit coordinate volume). With $\sqrt{|g|} = na^3b^4$ and $\delta S_m = \frac{1}{2}\int\sqrt{|g|}T_{\mu\nu}\delta g^{\mu\nu}$:

$$\varepsilon = \frac{W}{a^3b^4}, \qquad p_a = -\frac{\partial_a W}{3a^2b^4}, \qquad p_b = -\frac{\partial_b W}{4a^3b^3}, \qquad \frac{d}{dx_0}\frac{W}{a^3b^4} = -3H_a(\varepsilon + p_a) - 4H_b(\varepsilon + p_b),$$

the last identity being the chain rule $W' = \partial_a W\,a' + \partial_b W\,b'$. For one massless particle $W = \omega = \sqrt{k_a^2/a^2 + k_b^2/b^2}$, whence $\partial_a W = -k_a^2/(a^3\omega)$ and $\partial_b W = -k_b^2/(b^3\omega)$. The minisuperspace action $-\frac{1}{2\kappa}\int na^3b^4\sum_k \alpha_k L_k\,dx_0 - \int nW\,dx_0$ with the identities $\mathrm{EL}_n(na^3b^4L_k) = -2a^3b^4E^{(k)0}{}_0$, $\mathrm{EL}_a(na^3b^4L_k) = -6a^2b^4E^{(k)1}{}_1$, $\mathrm{EL}_b(na^3b^4L_k) = -8a^3b^3E^{(k)4}{}_4$ (at $n=1$) then gives

$$\mathrm{EL}_n = \frac{a^3b^4}{\kappa}\big(S_{00} - \kappa\varepsilon\big), \qquad \mathrm{EL}_a = \frac{3a^2b^4}{\kappa}\big(S_{11} + \kappa p_a\big), \qquad \mathrm{EL}_b = \frac{4a^3b^3}{\kappa}\big(S_{44} + \kappa p_b\big), \qquad S_{cc} = \sum_k \alpha_k E^{(k)c}{}_c.$$

The radion equation. With $\alpha_2 = \alpha_3 = 0$ the 11 and 44 equations are linear in H_a', H_b'; solving for H_b' and replacing $3H_a^2$ by the constraint $3H_a^2 = \kappa\varepsilon + \Lambda - 12H_aH_b - 6H_b^2$ gives

$$H_b' = -3H_aH_b - 4H_b^2 + \kappa\Big(\frac{\varepsilon}{6} - \frac{p_a}{2} + \frac{p_b}{3}\Big) + \frac{\Lambda}{3}, \qquad H_b = \beta'.$$

With the potential ($\varepsilon \to \varepsilon + U$, $p_a \to p_a - U$, $p_b \to p_b - U - bU'/4$) the right-hand side contains $\kappa(U/6 + U/2 - U/3 - bU'/12) = \kappa(U/3 - bU'/12)$; for $U = U_0 + \frac{K_U}{2}\beta^2$, $bU' = K_U\beta$, its linear part is $-(\kappa K_U/12)\beta$, so $m_r^2 = \kappa K_U/12$. **Static auxiliary space.** At $H_b = H_b' = 0$ the cubic tensor vanishes ($E^{(3)} \propto H_b$ in every component), the 00 and 11 equations give the brane Friedmann equations and the 44 equation gives

$$\frac{\kappa}{4}bU'(b) = \Lambda + \kappa U + \frac{2\kappa f_W}{V_\perp} + \frac{\kappa}{2}(\varepsilon - 3p_a) + \kappa p_b - 12\alpha_2 H_a^2(H_a' + H_a^2), \qquad 8\pi G_4 = \frac{\kappa}{V_\perp}, \qquad \Lambda_4 = \Lambda + \kappa U_0 + \frac{\kappa f_W}{V_\perp}.$$

Einstein frame. The identity $3(H_a + 2H_b)^2 - 6H_b^2 = 3(H_a^2 + 4H_aH_b + 2H_b^2) = E^{(1)0}{}_0$ turns the Einstein constraint into $3H_E^2 = 6\beta'^2 + \kappa\varepsilon$. With $\kappa = 8\pi G_8$, $8\pi G_4 = \kappa/V_\perp$, $M_{\mathrm{Pl}}^2 = V_\perp/\kappa$ and the energy density per brane volume $\rho = V_\perp\varepsilon$, this is $3M_{\mathrm{Pl}}^2H_E^2 = \frac{1}{2}\phi'^2 + \rho$ with $\phi = \sqrt{12}M_{\mathrm{Pl}}\beta$. A brane field sees $h = (b_0/b)^4 g_E$: in four dimensions $\sqrt{|h|}R[h] = (b_0/b)^4\sqrt{|g_E|}R[g_E]$ up to derivatives of β, and with the factor $V_\perp \propto b^4$ of the reduction the Einstein-frame metric g_E has the fixed Newton constant G_4. Since $\delta S = \frac{1}{2}\int\sqrt{|h|}T_{\mu\nu}\delta h^{\mu\nu} = -\frac{1}{2}\int\sqrt{|h|}T^{\mu\nu}\delta h_{\mu\nu}$ and $\delta h_{\mu\nu} = (e^{-4\beta} - 1)g_{E\mu\nu} \approx -4\beta h_{\mu\nu}$, the brane action changes by $2\beta\int\sqrt{|h|}T^{\mu}{}_{\mu}$: the coupling is $\mathcal{L}_{\phi W} = 2\beta T^{\mu}{}_{\mu} = (\phi/(\sqrt{3}M_{\mathrm{Pl}}))T^{\mu}{}_{\mu}$.

The vacuum curve. Divided by H_a^2 the vacuum constraint is $P_E(u) + XP_G(u) + \xi X^2 P_C(u) = 0$ with $X = H_a^2$, $u = H_b/H_a$ and

$$P_E = 3(2u^2+4u+1), \qquad P_G = -12u(u^3+12u^2+18u+4), \qquad P_C = 360u^3(3u+4).$$

For $u_1 < u < 0$ ($u_1 = -1+1/\sqrt{2}$, the root of P_E in $(-1,0)$): $P_E > 0$ and $P_C < 0$, so the product of the two roots of the quadratic, $P_E/(\xi P_C)$, is negative: exactly one root is positive, $X(u) = [P_G + \sqrt{P_G^2 - 4\xi P_C P_E}]/(2\xi|P_C|)$. The Kasner solutions $a \propto \tau^p$, $b \propto \tau^q$ have $H_a = p/\tau$, $H_b = q/\tau$, $u = q/p$; in Einstein gravity the constraint is $P_E(u) = 0$ and the 11 equation gives $3p+4q = 1$, hence $p_1 = 1/(3+4u_1) = (1+2\sqrt{2})/7$, $q_1 = (2-3\sqrt{2})/14$; in pure Gauss–Bonnet gravity $P_G(u) = 0$ (the root u_2 of $u^3+12u^2+18u+4$ in $(-\frac{1}{3},0)$) and $3p+4q = 3$.

The cubic regime. Inserting $H_a = 1/\tau + A_0 + A_1\tau + A_2\tau^2$, $H_b = h_0 + h_1\tau + h_2\tau^2 + h_3\tau^3$ (and the derivatives) into the three equations multiplied by τ^4 and collecting powers of τ: the order τ^0 of the 00 equation gives $h_0^2 = \alpha_2/(30\alpha_3)$; the next orders of the 00 and 44 equations give h_1, A_0, then h_2, A_1 (two linear equations at each order); the 11 equation is satisfied at these orders (a check). **The Einstein regime.** Substituting $u = u_1 + d_1 H_a^2 + d_2 H_a^4$ into the constraint and expanding in H_a gives $d_1 = \alpha_2(16-11\sqrt{2})/4$; then $p_{\rm eff} = -H_a^2/H_a' = p_1 + e_1 H_a^2 + \dots$ with $e_1 = \alpha_2(1+9\sqrt{2})/7$, and integrating $d\tau = -dH_a/H_a' = p_{\rm eff} dH_a/H_a^2$:

$$d\tau = \frac{dH_a}{H_a'} = -\frac{dH_a}{F} = -p_{\rm eff}\frac{dH_a}{H_a^2} \quad \Rightarrow \quad \tau = -\int \Big(\frac{p_1}{H_a^2} + e_1\Big) dH_a = \frac{p_1}{H_a} - e_1 H_a + s_1,$$

$$\tau(H_a) = \tau(u_2) + \int_{H_a}^{H_a(u_2)} \frac{dH}{F} = \tau(u_2) + \frac{p_1}{H_a} - \frac{p_1}{H_a(u_2)} + \int_{H_a}^{H_a(u_2)} \Big(\frac{1}{F} - \frac{p_1}{H^2}\Big) dH,$$

$$s_1 = \lim_{H_a\to 0}\Big(\tau - \frac{p_1}{H_a}\Big) = \tau(u_2) - \frac{p_1}{H_a(u_2)} + \int_0^{H_a(u_2)} \Big(\frac{1}{F} - \frac{p_1}{H^2}\Big) dH.$$

(The integrand of the last integral tends to e_1 at $H = 0$, so the integral converges; it is evaluated along the curve in the variable $u = u_1 + w^2$, and $\tau(u_2) = \int_{H_a(u_2)}^{\infty} dH/F$ in the variable $u = -v^2$.) **The Gauss–Bonnet regime.** $p_{\rm eff} = p_2 + f_1/H_a^2 + f_2\xi H_a^2$: f_1 and f_2 are the limits of $(p_{\rm eff}-p_2)H_a^2$ at $\xi = 0$ and of $(p_{\rm eff}-p_2)/(\xi H_a^2)$ at $H_a^2 \gg 1$, computed with 40-digit arithmetic at two points each (their agreement to 10^{-5} is a check). Integrating as above, $\tau = p_2/H_a + f_1/(3H_a^3) - f_2\xi H_a + s_2$.

The matter-dominated cubic regime. With $H_a = p/\tau$, $H_b = h$ finite and $\tau \to 0$, the leading terms are

$$\tau^3 \textstyle\sum \alpha_k E^{(k)0}{}_0 \to 48hp^3(30\alpha_3 h^2 - \alpha_2), \qquad \tau^3 \sum \alpha_k E^{(k)1}{}_1 \to 32hp^2(p-1)(18\alpha_3 h^2 - \alpha_2),$$
$$\tau^4 \textstyle\sum \alpha_k E^{(k)4}{}_4 \to p^3(p-1)(648\alpha_3 h^2 - 12\alpha_2).$$

The 44 equation (its right side $-\kappa p_b$ is of lower order) forces $h^2 = \alpha_2/(54\alpha_3)$ (for $p \neq 1$); the 11 equation $= -w_a \times$(00 equation) then gives $32hp^2(p-1)(18\alpha_3 h^2 - \alpha_2) = -w_a\, 48hp^3(30\alpha_3 h^2 - \alpha_2)$, i.e. with $18\alpha_3 h^2 - \alpha_2 = -\frac{2}{3}\alpha_2$ and $30\alpha_3 h^2 - \alpha_2 = -\frac{4}{9}\alpha_2$: $(p-1) = -w_a p$, $p = 1/(1+w_a)$; and $\kappa\varepsilon\tau^3 = 48|h|p^3 \cdot \frac{4}{9}\alpha_2 = \frac{64}{3}p^3\alpha_2|h|$ ($h < 0$).

Einstein gravity with a fluid. For $a = \tau^P$, $b = \tau^Q$, $\tau^2 E^{(1)}$ is constant and the three equations are

$$3(P^2+4PQ+2Q^2) = \mathcal{C}, \qquad 3P^2+8PQ-2P+10Q^2-4Q = -w_a\mathcal{C}, \qquad 3(2P^2+3PQ-P+2Q^2-Q) = -w_b\mathcal{C},$$

with conservation $\varepsilon \propto \tau^{-3P(1+w_a)-4Q(1+w_b)} = \tau^{-2}$; for $(w_a, w_b) = (0, \frac{1}{4})$: $P = 0$, $Q = \frac{2}{5}$, $\mathcal{C} = \frac{24}{25}$.

The tail of pair creation. $p_1 - q_1 = \frac{1+2\sqrt{2}}{7} - \frac{2-3\sqrt{2}}{14} = \frac{\sqrt{2}}{2}$ and $1 - p_1 = \frac{6-2\sqrt{2}}{7}$, so

$$\frac{2(p_1-q_1)}{1-p_1} = \frac{7\sqrt{2}}{6-2\sqrt{2}} = \frac{7\sqrt{2}(6+2\sqrt{2})}{28} = \frac{3\sqrt{2}+2}{2}.$$

The exponent follows from first-order adiabatic perturbation theory for the mode equation

$$i\,\frac{dc}{d\tau} = \omega\,(\cos\theta\,\sigma_1 + \sin\theta\,\sigma_2)\,c, \qquad \omega = K\sqrt{\tau^{-2p_1} + \tau^{-2q_1}}, \qquad \tan\theta = \tau^{p_1-q_1}:$$

$$\beta \approx \int_0^\infty \frac{1}{2}\frac{d\theta}{d\tau}\exp\Big(2i\int_0^\tau \omega(\tau_1)\,d\tau_1\Big)d\tau, \qquad \theta \approx \tau^{p_1-q_1}, \quad \omega \approx K\tau^{-p_1} \quad (\tau \to 0);$$
$$x = K\tau^{1-p_1}: \qquad \beta \approx K^{-\frac{p_1-q_1}{1-p_1}}\,\frac{p_1-q_1}{2(1-p_1)}\int_0^\infty x^{\frac{p_1-q_1}{1-p_1}-1} e^{2ix/(1-p_1)}dx,$$

so for large K (where the region $\tau \ll 1$ dominates) $|\beta|^2 \propto K^{-2(p_1-q_1)/(1-p_1)}$; the notebook and the Python program measure the exponent between $K = 8$ and $K = 16$. **The rate of pair creation near $\tau = 0$.** A mode becomes non-adiabatic when its physical energy is of order $1/\tau$. The physical number density of such modes in seven space dimensions is of order τ^{-7}, and the comoving volume is $a^3 b^4 = \tau^{3p_1+4q_1} = \tau$, so

$$\frac{dN_{\text{comoving}}}{d\ln\tau} \propto \tau^{-7}\tau^{3p_1+4q_1} = \tau^{-6};$$

the same exponent follows from the scaling variable: with $k_b/k_a = \tau_\times^{q_1-p_1}$ at the time $\tau_\times$ where the two terms are equal, $K = k_a\tau_\times^{1-p_1}$, and the modes with $K \lesssim 1$ (for which F is not suppressed by the power law; $F(0.3) \approx 0.14$) have $k_a \lesssim \tau_\times^{-(1-p_1)}$, $k_b = k_a\tau_\times^{-(p_1-q_1)}$; their number $\int d^3k_a\, d^4k_b$ per $\ln\tau_\times$ scales as $\tau_\times^{-7(1-p_1)-4(p_1-q_1)} = \tau_\times^{-(7-3p_1-4q_1)} = \tau_\times^{-6}$.

Radion capture. In a matter-dominated four-dimensional universe ($H_E = 2/(3\tau)$) the linearized radion equation is $\beta'' + 3H_E\beta' + m_r^2\beta = 0$, i.e. $\beta'' + \frac{2}{\tau}\beta' + m_r^2\beta = 0$, solved exactly by $\beta = A\sin(m_r\tau + \vartheta)/(m_r\tau)$ (substitute $\beta = v/\tau$: $v'' + m_r^2 v = 0$). The time origin of the Einstein frame is shifted, $\tau \to \tau - \tau_E$.

Radion decay. From $\mathcal{L}_{\phi W} = (\phi/(\sqrt{3}M_{\rm Pl}))T^\mu{}_\mu$ with $T^\mu{}_\mu \supset m_f\bar f f$, the amplitude for $\phi \to f\bar f$ is that of a scalar with Yukawa coupling $y = m_f/(\sqrt{3}M_{\rm Pl})$, $\Gamma = N_c y^2 m_r(1-4m_f^2/m_r^2)^{3/2}/(8\pi)$:

$$\Gamma(\phi\to f\bar f) = \frac{N_c m_f^2 m_r}{24\pi M_{\rm Pl}^2}\Big(1-\frac{4m_f^2}{m_r^2}\Big)^{3/2}, \qquad \Gamma(\phi\to gg) = \frac{b_{\rm QCD}^2\alpha_s^2 m_r^3}{96\pi^3 M_{\rm Pl}^2} \quad \Big(T^\mu{}_\mu \supset -\frac{b_{\rm QCD}\alpha_s}{8\pi}G^a_{\mu\nu}G^{a\mu\nu}\Big),$$

and the reheating temperature follows from $\Gamma = H = (\pi^2 g_*/90)^{1/2}T_{\rm RH}^2/M_{\rm Pl}$. **Kaluza–Klein channels.** The momenta of the torus are $\vec q = 2\pi\vec n/(b_0 L_c)$, $\vec n \in \mathbb{Z}^4$, one per volume $(2\pi)^4/V_\perp$ of momentum space; the four-ball of radius R has volume $\pi^2R^4/2$, so $N_{\rm KK}(|q|<R) = V_\perp\pi^2R^4/(2(2\pi)^4)$. **Normalcy temperature.** A brane plasma loses energy to bulk gravitons at the rate $\dot\rho/\rho \sim T^7/M_8^6$ (ADD); this is smaller than the expansion rate $H \sim T^2/M_{\rm Pl}$ for $T < T_*$, $T_*^5 = M_8^6/M_{\rm Pl}$ (an order-of-magnitude estimate).

Decay into the bulk. Reduced on the torus, a mode χ of dirac16complex with Kaluza–Klein momentum μ_n (at $b = b_0$) has the Jordan-frame mass $M_J = \sqrt{m^2 + \mu_n^2(b_0/b)^2}$. The brane metric is $h = \Omega^2 g_E$ with $\Omega^2 = (b_0/b)^4 = e^{-4\beta}$; with $\sqrt{|h|} = \Omega^4\sqrt{|g_E|}$ and the canonical rescaling $\chi = \Omega^{-3/2}\chi_E$ of a fermion, the mass term $\sqrt{|h|}M_J\bar\chi\chi = \sqrt{|g_E|}\,\Omega M_J\bar\chi_E\chi_E$, so

$$M_E(\beta) = e^{-2\beta}\sqrt{m^2+\mu_n^2e^{-2\beta}}, \qquad -\frac{\partial M_E}{\partial\beta}\Big|_0 = 2M_n + \frac{\mu_n^2}{M_n}, \qquad g_n = \frac{2M_n + \mu_n^2/M_n}{\sqrt{12}\,M_{\rm Pl}} \quad (\phi = \sqrt{12}M_{\rm Pl}\beta),$$

the diagonal coupling of the mass eigenstates (for $m = 0$ there is no other). The rate of $\phi \to \chi\bar\chi$ for one Dirac fermion with $-g\phi\bar\chi\chi$ is $g^2 m_r(1-4M^2/m_r^2)^{3/2}/(8\pi)$ (with $g = m_f/(\sqrt{3}M_{\rm Pl})$ it is the brane formula above); each level has four brane Dirac spinors (the 16 components), so for $m \ll \mu_n$, $\Gamma_n = 4\cdot\frac{3\mu_n^2}{4M_{\rm Pl}^2}\,m_r(1-4\mu_n^2/m_r^2)^{3/2}/(8\pi)$. In the continuum limit $\sum_n \to V_\perp\int d^4\mu/(2\pi)^4 = V_\perp\,2\pi^2\int\mu^3 d\mu/(2\pi)^4$ and, with $u = 2\mu/m_r$,

$$\Gamma_{\rm bulk} = \frac{3m_r}{8\pi M_{\rm Pl}^2}\frac{V_\perp}{8\pi^2}\int_0^{m_r/2}\mu^5\Big(1-\frac{4\mu^2}{m_r^2}\Big)^{3/2}d\mu = \frac{3m_r}{8\pi M_{\rm Pl}^2}\frac{V_\perp}{8\pi^2}\Big(\frac{m_r}{2}\Big)^6\int_0^1 u^5(1-u^2)^{3/2}du,$$

$$\int_0^1 u^5(1-u^2)^{3/2}du = \frac{B(3,\frac52)}{2} = \frac{8}{315}.$$

The exact sum uses Jacobi's four-square theorem: the number of $n \in \mathbb{Z}^4$ with $|n|^2 = k$ is $r_4(k) = 8\sum_{d|k,\,4\nmid d} d = 8\sigma(k) - 32\sigma(k/4)$.

$g_,g_{s}andt(T)$. For an ideal gas of species i with $x = m_i/T$, expanding the Bose/Fermi occupation $1/(e^{E/T}\mp 1) = \sum_{n\ge1}(\pm1)^{n+1}e^{-nE/T}$ and using $\int_m^\infty E\sqrt{E^2-m^2}e^{-nE/T}dE$-type integrals gives the Bessel series

$$\rho_i = \frac{g_iT^4}{2\pi^2}\sum_{n\ge1}(\pm1)^{n+1}\frac{3(nx)^2K_2(nx)+(nx)^3K_1(nx)}{n^4},$$

$$p_i = \frac{g_iT^4}{2\pi^2}\sum_{n\ge1}(\pm1)^{n+1}\frac{(nx)^2K_2(nx)}{n^4}, \qquad s_i = \frac{\rho_i+p_i}{T},$$

(400 terms in the notebook; the Python program integrates the momentum integrals instead). With sa^3 constant, $\ln a = -\ln T - \frac{1}{3}\ln g_{*s}$+const, and $dt = d\ln a/H_a$. On a step from $\ell_i = \ln a_i$ to ℓ_{i+1} with $1/H_a = e^{k(\ell-\ell_i)}/H_i$, $k = \ln(H_i/H_{i+1})/\Delta\ell$:

$$\int_{\ell_i}^{\ell_{i+1}} \frac{d\ell}{H_a} = \frac{e^{k\Delta\ell}-1}{kH_i} = \Delta\ell\,\frac{1/H_{i+1}-1/H_i}{\ln(H_i/H_{i+1})}, \qquad t(T_1) = \frac{1}{2H_a(T_1)} \text{ (radiation domination).}$$

The late universe. For radiation + matter, $H_0 t = \int_0^a \frac{x\,dx}{\sqrt{\Omega_r+\Omega_m x}} = \frac{2}{3\Omega_m^2}[(\Omega_m a - 2\Omega_r)\sqrt{\Omega_r+\Omega_m a} + 2\Omega_r^{3/2}]$; for matter + Λ, $H_0 t = \int_0^a \frac{\sqrt{x}\,dx}{\sqrt{\Omega_m+\Omega_\Lambda x^3}} = \frac{2}{3\sqrt{\Omega_\Lambda}}\mathrm{arcsinh}\big(\sqrt{\Omega_\Lambda/\Omega_m}\,a^{3/2}\big)$; both contain the matter-only solution $t_m = 2a^{3/2}/(3H_0\sqrt{\Omega_m})$, which the composite subtracts once. The same integral with $\Omega_r = 0$ inverted gives $a \propto \sinh^{2/3}(\frac{3}{2}H_\infty t)$ and $H_a = H_\infty \coth(\frac{3}{2}H_\infty t)$, $H_\infty = H_0\sqrt{\Omega_\Lambda}$.

The radion in de Sitter. Write the 00, 11 and 44 equations with $U = \frac{K_U}{2}\beta^2$ (in units $\kappa = 1$: $\kappa U = 6m_r^2\beta^2$, $\kappa b U'/4 = 3m_r^2\beta$), linearize about $(H_a, H_b, \beta) = (H, 0, \beta_{\rm eq})$ with Λ fixed by the 00 equation there, remove δH_a with the linearized constraint, and read off $\beta'' = -\gamma\beta' - m_{\rm eff}^2(\beta - \beta_{\rm eq})$:

$$\frac{\gamma}{3H} = \frac{1 - 16\alpha_2 H^2 + 48\alpha_2^2 H^4 - 48\alpha_3 H^4 + 512\alpha_2^3 H^6 + 192\alpha_2\alpha_3 H^6 - 1024\alpha_2^4 H^8}{(1 - 4\alpha_2 H^2 + 32\alpha_2^2 H^4)^2},$$
$$\frac{m_{\rm eff}^2}{m_r^2} = \frac{1}{1 - 4\alpha_2 H^2 + 32\alpha_2^2 H^4};$$

the equilibrium is the root of the 44 equation at $H_b = H_b' = 0$, $3m_r^2\beta_{\rm eq} = 3H^2 - 12\alpha_2 H^4 + O(\beta_{\rm eq}^2)$. For $a = b = e^{Hx_0}$ all three equations reduce to $21H^2 - 420\alpha_2 H^4 + 2520\alpha_3 H^6 = \Lambda$.

5. Verification

Evaluation of 7+1.nb: Wolfram 15.0.1 for Microsoft Windows (64-bit) (July 2, 2026), a fresh kernel (front-end kernel configuration Eval71); finished Wed 28 Oct 2026 08:57:47; 283.883 s; message cells 0; output cells 554. The checks of sections 20–23:

#	check	result	section
1	cubic Lovelock density in invariant form: L_3 == R^3 - 12 R Ric.Ric + 16 Ric^3 + 24 R^ac R^bd R_abcd + 3 R Riem.Riem - 24 R^ab R_a^cde R_bcde + 4 R^ab_cd R^cd_ef R^ef_ab - 8 R_a^c_b^d R_c^e_d^f R_e^a_f^b (exact, four random integer algebraic curvature tensors)	True	20.1
2	Gauss-Bonnet density in invariant form: L_2 == R^2 - 4 Ric.Ric + Riem.Riem (the same tensors, exact)	True	20.1
3	stabilizing potential: W = a^3 b^4 U(b) gives eps = U, pa = -U, pb = -U - (b/4) U'(b)	True	20.2
4	brane tension smeared over the torus (V_perp = b^4 Lc^4): eps = f_W/V_perp, pa = -f_W/V_perp, pb = 0	True	20.2
5	brane fluid rho_4 = N a^(-3(1+w)), smeared: eps = rho_4/V_perp, pa = w eps, pb = 0	True	20.2
6	massless dirac16complex particle (comoving momentum (k_a, k_b)): eps V = omega, pa V = k_a^2/(3 a^2 omega), pb V = k_b^2/(4 b^2 omega)	True	20.2
7	every such source is conserved: eps' + 3Ha(eps + pa) + 4Hb(eps + pb) == 0 for arbitrary W(a, b), a(x0), b(x0)	True	20.2
8	minisuperspace: EL_n == (a^3 b^4/kappa)(S00 - kappa eps), EL_a == (3 a^2 b^4/kappa)(S11 + kappa pa), EL_b == (4 a^3 b^3/kappa)(S44 + kappa pb) with S = Sum alpha_k E^(k) (identities of section 12.6)	True	20.2
9	radion equation of Einstein gravity: Hb' + 3 Ha Hb + 4 Hb^2 == kappa (eps/6 - pa/2 + pb/3) + Lambda/3 (constraint used)	True	20.3
10	radion mass: U = U0 + (K_U/2) beta^2, b U' = K_U beta gives the restoring term -(kappa K_U/12) beta, m_r^2 = kappa K_U/12	True	20.3
11	static auxiliary space: (kappa/4) b U' == Lambda + kappa U + 2 kappa f_W/V_perp + (kappa/2)(eps - 3pa) + kappa pb - 12 alpha2 Ha^2 (Ha' + Ha^2)	True	20.3
12	static auxiliary space: Ha' == -(kappa/2)(eps + pa) and E^(3) does not contribute (Hb = 0)	True	20.3
13	brane Friedmann equation: 3 Ha^2 == (kappa/V_perp) rho_4 + Lambda4 with Lambda4 = Lambda + kappa U0 + kappa f_W/V_perp, i.e. 8 Pi G4 = kappa/V_perp	True	20.3
14	Einstein-frame form of the constraint: 3 (Ha + 2 Hb)^2 == 6 Hb^2 + (E^(1)0_0 of section 12.6), i.e. 3 M_Pl^2 H_E^2 == phi'^2/2 + rho with phi = Sqrt[12] M_Pl beta	True	20.3

#	check	result	section
15	vacuum constraint in u = Hb/Ha, X = Ha^2 (alpha2 = 1, alpha3 = xi): P_E + X P_G + xi X^2 P_C == 0 with P_E = 3(2u^2 + 4u + 1), P_G = -12u(u^3 + 12u^2 + 18u + 4), P_C = 360 u^3 (3u + 4)	True	21.1
16	Kasner constants: P_E(u1) == 0, p1 == (1 + 2 Sqrt[2])/7, q1 == (2 - 3 Sqrt[2])/14; u2 in (-1/3, 0) with P_G(u2) == 0, 3 p2 + 4 q2 == 3	True	21.1
17	the explicit curve solves the vacuum constraint: residual < 10^-25 (relative) at u = u1 + 10^-6, -0.28, u2, -0.1, -10^-4 (xi = 10^-6, 40 digits)	True	21.1
18	the effective exponent along the curve: p_eff -> p1 (u -> u1), about p2 at u = u2 (within 10^-2), -> 1 (u -> 0)	True	21.1
19	cubic regime: Ha = 1/tau + A0 + A1 tau, Hb = h0 + h1 tau + h2 tau^2 solves the 00, 11, 44 equations through order tau^-1 (11: checked, not used)	True	21.2
20	cubic regime coefficients: h1 == alpha2/(16 alpha3) - 1/(32 alpha2), A0 == Sqrt[30](58 alpha2^2 - 45 alpha3)/(1920 alpha2^(3/2) alpha3^(1/2)), A1 and h2 as in the text	True	21.2
21	Einstein regime: u = u1 + d1 Ha^2, d1 == alpha2 (16 - 11 Sqrt[2])/4; p_eff = p1 + e1 Ha^2, e1 == alpha2 (1 + 9 Sqrt[2])/7	True	21.2
22	Gauss-Bonnet regime: p_eff = p2 + f1/Ha^2 + f2 xi Ha^2 with f1 = -0.69887346..., f2 = -0.7853407... (differences give the same values at Ha = 10^4 and 10^-6 xi Ha^2 to 10^-5)	True	21.2
23	cubic-series data at tau = 10^-8 satisfy the vacuum constraint to < 10^-12 (relative to 48 Ha^3 |Hb|)	True	21.2
24	the ODE solution agrees with the exact quadrature: tau at Ha(u2) to 10^-10, and tau - p1/Ha + e1 Ha at Ha = 3 10^-3 equals s1 to 10^-6	True	21.2
25	regime approximations: cubic error < 10^-5 for Ha >= 10^4, Gauss-Bonnet < 10^-3 for 3 <= Ha <= 300, Einstein < 10^-3 for Ha <= 0.1	True	21.2
26	matter-dominated cubic regime: p_eff -> 3/4, Hb -> -Sqrt[alpha2/(54 alpha3)], kappa eps tau^3 -> 9 alpha2 |h*|, alpha2 E2 : alpha3 E3 : kappa eps -> 9/4 : -5/4 : 1 (Ha = 10^15)	True	21.3
27	coupled: where the gas is a small fraction the effective exponents are those of the vacuum (1, p2, p1 within 10^-2)	True	21.3
28	without the stabilizing potential the gas-dominated Einstein regime decompactifies: tau Ha -> 0, tau Hb -> 2/5, pb/eps -> 1/4 (Ha = 10^-14)	True	21.3
29	the gas fraction grows like tau^(2 - 3 p1 - 5 q1) = tau^1.160 in the Einstein regime (to 2 10^-2)	True	21.3
30	Einstein + fluid power laws a ~ tau^P, b ~ tau^Q: (wa, wb) = (0, 1/4) -> (0, 2/5); (1/3, 0) -> (1/2, 0); (1/7, 1/7) -> (1/4, 1/4) (tau^2 E^(1) from EKas of section 14.1)	True	21.3
31	matter-dominated cubic regime (any fluid, b finite): leading 44 term p^3 (p - 1)(648 alpha3 h^2 - 12 alpha2); with h^2 = alpha2/(54 alpha3): p = 1/(1 + w_a) and kappa eps tau^3 = (64/3) p^3 alpha2 |h|	True	21.3
32	Einstein-Kasner pair creation: F(1) == 0.0107493 (section 14.3) and the tail F ~ K^(-(3 Sqrt[2] + 2)/2) (exponent between K = 8 and 16 to 2 10^-2)	True	21.4
33	brane plasma at T = M8, H = M8: fraction (Pi^2 g*/30) M8^4/(3 M_Pl^2 M8^2) < 10^-27	True	21.4
34	capture: the radion oscillates like a four-dimensional massive field in a matter-dominated universe: with tau_E from <H_E> = 2/(3 (tau - tau_E)) over the last period, beta = A sin(m_r tau + vartheta)/(m_r (tau - tau_E)) to 2 10^-2, <H_E>(tau - tau_E) = 2/3 to 10^-2 at tau = 10^5 and 2 10^5, the gas share constant to 2 10^-2	True	22.1
35	through brane fields alone a 10 GeV radion would decay far too late: T_RH < 10^-6 GeV (nucleosynthesis needs 4 10^-3 GeV); 10^13 Kaluza-Klein channels below m_r/2	True	22.2
36	the number of lattice vectors with |n|^2 = k is r_4(k) = 8 sigma(k) - 32 sigma(k/4) (direct count for k <= 60)	True	22.2
37	the radion decays into the bulk: the lattice sum over the Kaluza-Klein modes of dirac16complex (m << m_r/2) equals its continuum limit to 10^-6, the number of momenta to 10^-5, and Gamma_bulk/Gamma_brane > 10^12	True	22.2
38	g* limits: 106.75 above the electroweak scale (to 0.1), 10.75 near 3 MeV (to 0.1), 2 + (21/4)(4/11)^(4/3)(3.044/3) = 3.3855 after e+e- annihilation (to 10^-3)	True	22.3
39	the time-temperature quadrature is converged: halving the grid changes t(10 keV) by less than 10^-5	True	22.3
40	radiation era: t(1 MeV) = 0.74 s within 5% (1/(2 Ha) with g* = 10.75)	True	22.3
41	age of the universe from the brane Friedmann equation: 13.79 Gyr (Planck 2018: 13.787 +- 0.020)	True	22.4
42	composite analytic solution t(a) = t_rm(a) + t_mLambda(a) - t_m(a) accurate to < 3 10^-4 for 10^-8 <= a <= 10^0.5	True	22.4
43	far future: Ha(t) = H_inf Coth[3 H_inf t/2] agrees with the full brane solution at a = 3 and a = 10 to 10^-4	True	23.1
44	radion in de Sitter: gamma/(3H) and m_eff^2/m_r^2 as in the text (beta_eq -> 0); gamma = 3H and m_eff^2 = m_r^2 for alpha2 = alpha3 = 0	True	23.2
45	radion equilibrium in de Sitter: 3 m_r^2 beta_eq == 3 H^2 - 12 alpha2 H^4 + O(beta_eq^2)	True	23.2

#	check	result	section
46	the radion relaxes to the equilibrium of the two algebraic equations: beta(3000) to 10^-6, Ha(3000) to 10^-8 (relative)	True	23.2
47	eight-dimensional de Sitter (no stabilizer): all three equations read 21 H^2 - 420 alpha2 H^4 + 2520 alpha3 H^6 == Lambda	True	23.2
48	the shares of the four terms of the 00 equation: Einstein > 0.99 at Ha = 10^-2, Gauss-Bonnet > 0.9 at Ha = 30, Gauss-Bonnet and cubic about 1/2 each at Ha = 3 10^5, gas > 0.1 at Ha = 10^15	True	23.3

The independent Python program tools/ref_coupled.py — its boolean checks:

#	check	result
1	stabilizing potential: S_U = -Int sqrt|g| U(b) gives eps = U, pa = -U, pb = -U - (b/4) U'(b)	True
2	brane tension, smeared over the torus (delta^4(y) -> 1/Lc^4): eps = f_W/V_perp, pa = -f_W/V_perp, pb = 0, V_perp = b^4 Lc^4	True
3	brane fluid (Standard Model plasma, rho_4 = N_W a^(-3(1+w))), smeared: eps = rho_4/V_perp, pa = w eps, pb = 0	True
4	massless dirac16complex particle (comoving momenta k_a, k_b): eps V = omega, pa V = k_a^2/(3 a^2 omega), pb V = k_b^2/(4 b^2 omega), V = a^3 b^4	True
5	every such matter is conserved: eps' + 3Ha(eps + pa) + 4Hb(eps + pb) = 0 for arbitrary W(a, b), a(x0), b(x0)	True
6	field equations from S_grav + S_m: EL_n = (a^3 b^4/kappa)(S00 - kappa eps), EL_a = (3a^2 b^4/kappa)(S11 + kappa pa), EL_b = (4a^3 b^3/kappa)(S44 + kappa pb), S = Sum alpha_k E^(k) (minisuperspace, lapse n = 1)	True
7	radion equation (Einstein): Hb' + 3 Ha Hb + 4 Hb^2 = kappa (eps/6 - pa/2 + pb/3) + Lambda/3 (Hb = beta', exact)	True
8	radion mass: with U = U0 + (K/2) ln^2(b/b0) the restoring term is -m_r^2 beta, m_r^2 = kappa K/12	True
9	static auxiliary space: (kappa/4) b U'(b) = Lambda + kappa U + 2 kappa f_W/V_perp + (kappa/2)(eps - 3pa) + kappa pb - 12 alpha2 Ha^2 (Ha' + Ha^2)	True
10	static auxiliary space: E^(3) does not contribute, and Ha' = -(kappa/2)(eps + pa) (brane Friedmann acceleration)	True
11	four-dimensional Friedmann equation on the brane (b static): 3 Ha^2 = 8 pi G4 rho4 + Lambda4 with 8 pi G4 = kappa/V_perp and Lambda4 = Lambda + kappa U0 + kappa f_W/V_perp	True
12	cubic Lovelock density: L_3 = R^3 - 12 R Ric.Ric + 16 Ric^3 + 24 R^ac R^bd R_abcd + 3 R Riem.Riem - 24 R^ab R_a^cde R_bcde + 4 R^ab_cd R^cd_ef R^ef_ab - 8 R_a^c_b^d R_c^e_d^f R_e^a_f^b (exact, four random integer algebraic curvature tensors)	True
13	Gauss-Bonnet density: L_2 = R^2 - 4 Ric.Ric + Riem.Riem (same tensors)	True
14	vacuum constraint in u = Hb/Ha, X = Ha^2 (alpha2 = 1, alpha3 = xi): E(u) + X G(u) + xi X^2 K(u) = 0 with E = 3(2u^2 + 4u + 1), G = -12u(u^3 + 12u^2 + 18u + 4), K = 360 u^3 (3u + 4)	True
15	Einstein regime: u1 = Hb/Ha = -1 + 1/sqrt(2), root of E(u); Kasner p1 = 1/(3 + 4u1) = (1 + 2 sqrt 2)/7, q1 = u1 p1 = (2 - 3 sqrt 2)/14	True
16	Gauss-Bonnet regime: u2 = Hb/Ha, root of u^3 + 12u^2 + 18u + 4 in (-1/3, 0); Kasner p2 = 3/(3 + 4u2), q2 = u2 p2 (3p + 4q = 3)	True
17	exact vacuum solution: for every u in (u1, 0) the curve Ha = sqrt(X(u)), Hb = u Ha, X(u) = [G + sqrt(G^2 - 4 xi K E)]/(2 xi |K|), solves the constraint (residual < 1e-20 relative)	True
18	the effective exponent goes from p1 (Einstein, u -> u1) through about p2 (Gauss-Bonnet, u = u2) to 1 (cubic, u -> 0)	True
19	cubic regime (tau -> 0): Ha = 1/tau + A0 + A1 tau, Hb = h0 + h1 tau + h2 tau^2 solves all three equations to the orders used; h0 = -sqrt(alpha2/(30 alpha3))	True
20	cubic regime: h1 = alpha2/(16 alpha3) - 1/(32 alpha2), A0 = sqrt(30)(58 alpha2^2 - 45 alpha3)/(1920 alpha2^(3/2) alpha3^(1/2))	True
21	cubic regime: A1 = -3647 alpha2/(92160 alpha3) - 223/(6144 alpha2) + 645 alpha3/(8192 alpha2^3), h2 = sqrt(30)(-1212 alpha2^4 + 1180 alpha2^2 alpha3 + 225 alpha3^2)/(307200 alpha2^(5/2) alpha3^(3/2))	True
22	Einstein regime (Ha -> 0): u = u1 + d1 Ha^2 + ..., d1 = alpha2 (16 - 11 sqrt 2)/4; p_eff = p1 + e1 Ha^2 + ..., e1 = alpha2 (1 + 9 sqrt 2)/7	True
23	cubic-series initial data at tau = 1e-8 satisfy the constraint to < 1e-12 (relative to 48 alpha2 Ha^3 |Hb|)	True
24	the ODE solution and the quadrature agree: tau at Ha(u2) to 1e-10, s1 (ODE at Ha = 3e-3) to 1e-6	True
25	regime approximations: cubic error < 1e-5 for Ha >= 1e4, Gauss-Bonnet < 1e-3 for 3 <= Ha <= 300, Einstein < 1e-3 for Ha <= 0.1	True
26	shares: Einstein > 0.99 at Ha = 1e-2, Gauss-Bonnet > 0.9 at Ha = 30, Gauss-Bonnet and cubic about 1/2 each at Ha = 3e5 (to 1e-2), gas > 0.1 at Ha = 1e15	True
27	coupled: where the gas is a small fraction, the effective exponents are those of the vacuum solution (1, p2, p1 within 1e-2)	True

#	check	result
28	without a stabilizing potential the gas-dominated Einstein regime decompactifies: a -> const (tau Ha -> 0), b ~ tau^(2/5) (tau Hb -> 2/5), pb/eps -> 1/4	True
29	gas fraction grows like tau^(2 - 3p1 - 5q1) = tau^1.160 in the Einstein regime (to 2e-2)	True
30	Einstein + fluid power laws a ~ tau^P, b ~ tau^Q: (wa, wb) = (0, 1/4) -> (P, Q) = (0, 2/5) [decompactification]; (1/3, 0) -> (1/2, 0) [brane radiation, b frozen]; (1/7, 1/7) -> (1/4, 1/4) [isotropic]	True
31	matter-dominated cubic regime (any fluid pa = w_a eps, b finite): leading 44 equation p^3 (p - 1)(648 alpha3 h^2 - 12 alpha2) = 0 -> h^2 = alpha2/(54 alpha3); 11/00 -> p = 1/(1 + w_a); kappa eps tau^3 = (64/3) p^3 alpha2 |h|	True
32	pair creation has a power-law tail F ~ K^(-2(p1 - q1)/(1 - p1)) = K^(-(3 sqrt 2 + 2)/2) = K^(-3.121) (to 2e-2): it comes from the singularity, so the number of created pairs is dominated by the earliest times	True
33	F(1) agrees with the value of section 14.3 of 7+1.nb (|beta|^2 for (|p|, |q|) = (1, 1))	True
34	the Standard Model plasma on the brane (T <= M8) is gravitationally negligible in the bulk while H ~ M8: fraction < 1e-27	True
35	capture: the radion oscillates as a four-dimensional massive field in a matter-dominated universe: j0 form to 2e-2, <H_E>(tau - tau_E) -> 2/3 to 1e-2, the gas fraction stays constant to 2e-2	True
36	four-dimensional (Einstein-frame) form of the 00 equation: 3 (Ha + 2Hb)^2 = 6 Hb^2 + kappa eps_total, i.e. 3 H_E^2 M_Pl^2 = (1/2) phi'^2 + rho with phi = sqrt(12) M_Pl ln(b/b0)	True
37	through brane fields alone a radion of 10 GeV would decay far too late (T_RH < 1 keV < 4 MeV needed by nucleosynthesis): the moduli problem of the minimal model	True
38	Jacobi's four-square formula r_4(k) = 8 sigma(k) - 32 sigma(k/4) equals the direct count of lattice vectors for 1 <= k <= 60	True
39	the radion decays into the bulk: lattice sum = continuum limit to 1e-6, number of momenta = N_KK to 1e-5, Gamma_bulk/Gamma_brane > 1e12	True
40	g* limits: 106.75 above the electroweak scale (to 0.1), 10.75 near 3 MeV (to 0.1), and after e+e- annihilation 2 + (21/4)(4/11)^(4/3)(N_eff/3) = 3.3855 (to 1e-3)	True
41	radiation era: the t(T) quadrature is converged (halving the grid changes t(10 keV) by < 1e-5)	True
42	radiation era: t(1 MeV) = 0.74 s within 5% (standard value 1/(2H) with g* = 10.75)	True
43	age of the universe 13.79 Gyr (Planck 2018: 13.787 +- 0.020)	True
44	the composite analytic solution (exact radiation+matter, exact matter+Lambda, minus their common matter part) is accurate to < 3e-4 at all times	True
45	far future: H(t) = H_inf coth(3 H_inf t/2) (exact for matter + Lambda) agrees with the full solution at a = 3, 10 to 1e-4	True
46	far future: the radion obeys beta'' + gamma beta' + m_eff^2 beta = 0 around de Sitter (constraint used for delta Ha); alpha2 = alpha3 = 0: gamma = 3H, m_eff^2 = m_r^2 at beta_eq -> 0	True
47	far future: equilibrium of the radion in de Sitter: 3 m_r^2 beta_eq = 3H^2 - 12 alpha2 H^4 (+ O(beta_eq^2))	True
48	eight-dimensional de Sitter (no stabilizer): 21 H^2 - 420 alpha2 H^4 + 2520 alpha3 H^6 = Lambda (all three equations)	True
49	far future: the radion relaxes to its shifted equilibrium (to 1e-6 relative) and Ha to H_eq (to 1e-8)	True

Numbers printed by both programs (largest relative difference per group, and the tolerance used by tools/verify_7p1_nb.py; numbers read off long numerical integrations, such as the two quantities τH_a and p_a/ε that tend to zero in the gas-dominated late regime, agree less closely than closed forms):

quantity	values	max. relative difference	tolerance
Kasner constants $u_1, u_2, p_1, q_1, p_2, q_2$	6	0	1×10^{-14}
p_{eff} at $u \to u_1$, $u = u_2$, $u \to 0$	3	0	1×10^{-10}
cubic-regime coefficients h_0, h_1, h_2, A_0, A_1	5	0	1×10^{-13}
Gauss–Bonnet coefficients f_1, f_2	2	1.6×10^{-16}	1×10^{-9}
$\tau(u_2)$, $H_a(u_2)$, s_1 (quadrature)	3	6.4×10^{-16}	1×10^{-12}
s_2, $H_{a,\text{mid}}$	2	1.6×10^{-11}	1×10^{-8}
maximal errors of the regime formulas	4	5.3×10^{-8}	0.001
crossover times $\tau(H_a = 10^3)$, $\tau(H_a = 1)$	2	4.6×10^{-13}	1×10^{-9}
coupled, near the singularity	5	1.4×10^{-13}	1×10^{-8}
coupled: (p_{eff}, gas fraction) in the three Lovelock regimes	6	6.0×10^{-12}	1×10^{-6}
coupled, gas-dominated late regime	4	0.00032	0.002
growth exponent of the gas fraction (Einstein regime)	2	1.5×10^{-11}	1×10^{-8}
pair creation $F(1), F(8), F(16)$ and the tail exponent	4	1.2×10^{-9}	1×10^{-6}

quantity	values	max. relative difference	tolerance
brane plasma fraction at $T = H = M_8$	1	2.3×10^{-16}	1×10^{-12}
capture: start τ, β	2	3.8×10^{-12}	1×10^{-8}
capture: A, ϑ, τ_E, residual	4	3.9×10^{-9}	1×10^{-6}
capture: $\langle H_E \rangle (\tau - \tau_E)$	2	9.2×10^{-12}	1×10^{-8}
capture: gas share, radion share	3	1.5×10^{-10}	1×10^{-8}
radion decay: Γ, lifetime, T_{RH}	3	3.7×10^{-16}	1×10^{-12}
Kaluza–Klein momenta below $m_r/2$	1	1.7×10^{-16}	1×10^{-12}
radion decay into the bulk: lattice sum, continuum limit, lifetime, ratio	4	8.6×10^{-16}	1×10^{-9}
$V_\perp$, $\ell_\perp$, $2\pi/\ell_\perp$	3	2.2×10^{-16}	1×10^{-12}
normalcy temperature T_*	1	0	1×10^{-12}
$g_*(T)$, $g_{*s}(T)$	18	5.5×10^{-9}	1×10^{-6}
$t(T)$	6	2.7×10^{-9}	1×10^{-6}
$1/(2H_a)$ divided by $t(T)$	6	7.6×10^{-11}	1×10^{-6}
Ω_r, Ω_Λ, z_{eq}, age	4	1.7×10^{-15}	1×10^{-9}
maximal error of the composite late-time solution	1	6.6×10^{-12}	1×10^{-6}
radion displacement today	2	4.1×10^{-16}	1×10^{-12}
$\alpha_2 H_0^2$, $\alpha_3 H_0^4$	2	1.0×10^{-15}	1×10^{-12}
H_∞, Gibbons–Hawking temperature	3	3.2×10^{-16}	1×10^{-12}
$2\pi M / H_\infty$	2	2.3×10^{-16}	1×10^{-12}
de Sitter radion test	4	8.5×10^{-16}	1×10^{-9}
shares of the four terms of the 00 equation (section 9)	16	1.0×10^{-11}	1×10^{-6}

SHA-256 of the files (at the time this document was generated):

```
7+1.nb                          41d252f9852f210c228f45e1ac956d2d7ef26faeb0062a5880280e1aae0ef
    ...fc7
tools/make_7p1_nb.py            996d26c9552cd44902560af16a1683fa71457261f18fefaeddae87bd99a2b
    ...144
tools/nb_part_gravity.py        c4caf2ad5c932e8f603341edade5f4b98fa784feb05906c051eab7c884cd6
    ...8f8
tools/nb_part_ks_gravity.py     a656c0ad040b4576b75b17e029545109e531c8e656ad99bfdb5cba946e967
    ...772
tools/nb_part_sm.py             43e92beed57453a4b57931de6d603ac6c218efba6e4005133b69b8054a975
    ...c5b
tools/nb_part_coupled.py        e630521d64132e71afc861114f699eb06454557cf14a57ad8ec1eef96247a
    ...ae0
tools/nb_part_dark.py           10ad8fb8ae25fa07bbee758c653995acd4e2c80733059a8b3648bd1593e46
    ...db0
tools/nb_part_torus_brane.py    fb904ff8d85b904c55b3b2a8f60fb0cb047f1ccf1b98db3364092f2bae0b2
    ...162
tools/nb_part_radion.py         53c5604707a3e22228c840ee0a2d2125616f9410f75a5535041dfaf18f497
    ...c00
tools/lovelock71.py             d93ecbd0f82adb972218d59dfe32999a4af944945ca9758607f3a633e1af9
    ...5a1
tools/ref_gravity.py            f07cf8adaff585af89c87d19532f1aaa7db606ad88ae90483b2d1f41341ed
    ...f36
tools/ref_ks_gravity.py         b6488ef24b1c16aacd8ff6cf6a5a1ba3f316e082449ad97480022d1a26383
    ...27f
tools/ref_sm.py                 f3f80806893f42bbf166e1f1de896ffd611dbfe847685da8e3ee367e9cde6
    ...aba
tools/ref_coupled.py            695dbd9375753653472d9c37502f0067c4a64fe01a64ba8c05e8b2c5da85a
    ...a58
tools/ref_dark.py               1600e6b57ab0383b4f2b664e2796d3864c9afa77a9ed59f007f66e1139453
    ...ff5
tools/ref_torus_brane.py        1f623656ab0f5c8ee43cb310f7bfd580cbba5a7385da2963adaae4d2f8bda
    ...af7
tools/ref_radion.py             e9d47176b9f2d2232a84b3b82188bd3b0a19cde3dbe1ba013c1fe5abff634
    ...1ce
tools/ref71.py                  1a7a06eb0add3f459611776dc9c474a5bb959d947a9db949db4612b1ab943
    ...4e0
```

```
tools/ref44.py                      ab100c3ebcc055fbe59c132a02f80ecb17630e934e99d59f88e919ec8064f
    ...af5
```

Output of tools/ref_coupled.py (independent checks of sections 20-23; exit code 0) (complete):

```
stabilizing potential: S_U = -Int sqrt|g| U(b) gives eps = U, pa = -U, pb = -U - (b/4) U'(b) :
    ...True
brane tension, smeared over the torus (delta^4(y) -> 1/Lc^4): eps = f_W/V_perp, pa =
    ...-f_W/V_perp, pb = 0, V_perp = b^4 Lc^4 : True
brane fluid (Standard Model plasma, rho_4 = N_W a^(-3(1+w))), smeared: eps = rho_4/V_perp, pa =
    ...w eps, pb = 0 : True
massless dirac16complex particle (comoving momenta k_a, k_b): eps V = omega, pa V = k_a^2/(3
    ...a^2 omega), pb V = k_b^2/(4 b^2 omega), V = a^3 b^4 : True
every such matter is conserved: eps' + 3Ha(eps + pa) + 4Hb(eps + pb) = 0 for arbitrary W(a, b),
    ...a(x0), b(x0) : True
field equations from S_grav + S_m: EL_n = (a^3 b^4/kappa)(S00 - kappa eps), EL_a = (3a^2
    ...b^4/kappa)(S11 + kappa pa), EL_b = (4a^3 b^3/kappa)(S44 + kappa pb), S = Sum alpha_k
    ...E^(k) (minisuperspace, lapse n = 1) : True
radion equation (Einstein): Hb' + 3 Ha Hb + 4 Hb^2 = kappa (eps/6 - pa/2 + pb/3) + Lambda/3 (Hb
    ...= beta', exact) : True
radion mass: with U = U0 + (K/2) ln^2(b/b0) the restoring term is -m_r^2 beta, m_r^2 = kappa
    ...K/12 : True
static auxiliary space: (kappa/4) b U'(b) = Lambda + kappa U + 2 kappa f_W/V_perp +
    ...(kappa/2)(eps - 3pa) + kappa pb - 12 alpha2 Ha^2 (Ha' + Ha^2) : True
static auxiliary space: E^(3) does not contribute, and Ha' = -(kappa/2)(eps + pa) (brane
    ...Friedmann acceleration) : True
four-dimensional Friedmann equation on the brane (b static): 3 Ha^2 = 8 pi G4 rho4 + Lambda4
    ...with 8 pi G4 = kappa/V_perp and Lambda4 = Lambda + kappa U0 + kappa f_W/V_perp : True
cubic Lovelock density: L_3 = R^3 - 12 R Ric.Ric + 16 Ric^3 + 24 R^ac R^bd R_abcd + 3 R
    ...Riem.Riem - 24 R^ab R_a^cde R_bcde + 4 R^ab_cd R^cd_ef R^ef_ab - 8 R_a^c_b^d R_c^e_d^f
    ...R_e^a_f^b (exact, four random integer algebraic curvature tensors) : True
Gauss-Bonnet density: L_2 = R^2 - 4 Ric.Ric + Riem.Riem (same tensors) : True
vacuum constraint in u = Hb/Ha, X = Ha^2 (alpha2 = 1, alpha3 = xi): E(u) + X G(u) + xi X^2 K(u)
    ...= 0 with E = 3(2u^2 + 4u + 1), G = -12u(u^3 + 12u^2 + 18u + 4), K = 360 u^3 (3u + 4) :
    ...True
Einstein regime: u1 = Hb/Ha = -1 + 1/sqrt(2), root of E(u); Kasner p1 = 1/(3 + 4u1) = (1 + 2
    ...sqrt 2)/7, q1 = u1 p1 = (2 - 3 sqrt 2)/14 : True
Gauss-Bonnet regime: u2 = Hb/Ha, root of u^3 + 12u^2 + 18u + 4 in (-1/3, 0); Kasner p2 = 3/(3 +
    ...4u2), q2 = u2 p2 (3p + 4q = 3) : True
NUM u1, u2, p1, q1, p2, q2 : (-0.2928932188134525, -0.26958419031043745, 0.5469181606780271,
    ...-0.16018862050852037, 1.5611476243561566, -0.4208607182671175)
exact vacuum solution: for every u in (u1, 0) the curve Ha = sqrt(X(u)), Hb = u Ha, X(u) = [G +
    ...sqrt(G^2 - 4 xi K E)]/(2 xi |K|), solves the constraint (residual < 1e-20 relative) :
    ...True
NUM effective Kasner exponent p_eff = -Ha^2/Ha' along the curve at u -> u1, u = u2, u -> 0 (xi
    ...= 1e-6) : (0.5469181624462031, 1.553736173053791, 1.0000018125008945)
the effective exponent goes from p1 (Einstein, u -> u1) through about p2 (Gauss-Bonnet, u = u2)
    ...to 1 (cubic, u -> 0) : True
cubic regime (tau -> 0): Ha = 1/tau + A0 + A1 tau, Hb = h0 + h1 tau + h2 tau^2 solves all three
    ...equations to the orders used; h0 = -sqrt(alpha2/(30 alpha3)) : True
cubic regime: h1 = alpha2/(16 alpha3) - 1/(32 alpha2), A0 = sqrt(30)(58 alpha2^2 - 45
    ...alpha3)/(1920 alpha2^(3/2) alpha3^(1/2)) : True
cubic regime: A1 = -3647 alpha2/(92160 alpha3) - 223/(6144 alpha2) + 645 alpha3/(8192
    ...alpha2^3), h2 = sqrt(30)(-1212 alpha2^4 + 1180 alpha2^2 alpha3 + 225 alpha3^2)/(307200
    ...alpha2^(5/2) alpha3^(3/2)) : True
NUM cubic-regime coefficients (xi = 1e-6): h0, h1, h2, A0, A1 : (-182.57418583505537,
    ...62499.96875, -21609345.4877448, 165.4577275405445, -39572.51893438307)
Einstein regime (Ha -> 0): u = u1 + d1 Ha^2 + ..., d1 = alpha2 (16 - 11 sqrt 2)/4; p_eff = p1 +
    ...e1 Ha^2 + ..., e1 = alpha2 (1 + 9 sqrt 2)/7 : True
NUM Gauss-Bonnet regime: p_eff = p2 + f1/Ha^2 + f2 xi Ha^2 + ..., f1, f2 (alpha2 = 1) :
    ...(-0.6988734676355542, -0.7853407141875988)
cubic-series initial data at tau = 1e-8 satisfy the constraint to < 1e-12 (relative to 48
    ...alpha2 Ha^3 |Hb|) : True
```

```
NUM exact quadrature: tau(u2), Ha(u2), s1 = tau(u2) - p1/Ha(u2) + Int_0^Ha(u2) (1/F - p1/H^2)
    ...dH : (0.15935497980477448, 9.721560116243376, 2.0795884930371127)
the ODE solution and the quadrature agree: tau at Ha(u2) to 1e-10, s1 (ODE at Ha = 3e-3) to
    ...1e-6 : True
NUM time shifts: tau = p1/Ha - e1 Ha + s1 (Einstein), tau = p2/Ha + f1/(3 Ha^3) - f2 xi Ha + s2
    ...(Gauss-Bonnet, at Ha_mid = (f1/(f2 xi))^(1/4)): s1, s2, Ha_mid : (2.0795884930371127,
    ...np.float64(-0.0009864616381799337), 30.713910188158728)
NUM accuracy (max relative error of tau(Ha)): cubic for Ha >= 1e4, Gauss-Bonnet for 3 <= Ha <=
    ...300, Einstein for Ha <= 0.1 : (np.float64(6.651897785481253e-06),
    ...np.float64(0.0008521318063101635), np.float64(0.00032101971624165504))
regime approximations: cubic error < 1e-5 for Ha >= 1e4, Gauss-Bonnet < 1e-3 for 3 <= Ha <=
    ...300, Einstein < 1e-3 for Ha <= 0.1 : True
NUM worst error of the best regime approximation over 1e-3.5 <= Ha <= 1e8 (in the crossover
    ...windows) : 0.10466193313053529
NUM vacuum crossover times: tau at Ha = 1e3 (cubic -> GB) and Ha = 1 (GB -> Einstein) :
    ...(np.float64(0.0011471184489938008), np.float64(1.3947137801608218))
NUM coupled, near the singularity (Ha = 1e15): p_eff, Hb, kappa eps tau^3, alpha2 E2/(kappa
    ...eps), alpha3 E3/(kappa eps) : (np.float64(0.7500003270775101),
    ...np.float64(-136.08292174325447), np.float64(1224.7435263138136),
    ...np.float64(2.250006541529109), np.float64(-1.2500065415291277))
matter-dominated cubic regime: p_eff -> 3/4, Hb -> -sqrt(alpha2/(54 alpha3)), kappa eps tau^3
    ...-> 9 alpha2 |h*|, alpha2 E2 : alpha3 E3 : kappa eps -> 9/4 : -5/4 : 1 (to 1e-3) : True
NUM coupled: (p_eff, gas fraction kappa eps/(S1 + S2 + S3 + kappa eps)) at Ha = 3e5 (vacuum
    ...cubic), 30 (Gauss-Bonnet), 1e-2 (Einstein) : ((np.float64(0.9993182542368277),
    ...np.float64(0.0011887681564502635)), (np.float64(1.5596654174306679),
    ...np.float64(1.8450335287549852e-07)), (np.float64(0.5470861129173028),
    ...np.float64(2.8320201318030544e-05)))
NUM shares (Einstein S1, Gauss-Bonnet S2, cubic S3, gas kappa eps) of the 00 equation at Ha =
    ...3e5, 30, 1e-2, 1e15 : ((np.float64(5.730063146835421e-10),
    ...np.float64(0.5011389710000743), np.float64(0.49767226027046907),
    ...np.float64(0.0011887681564502635)), (np.float64(0.00023534899632421242),
    ...np.float64(0.9988000625274751), np.float64(0.0009644039728478138),
    ...np.float64(1.8450335287549852e-07)), (np.float64(0.99945569394049),
    ...np.float64(0.0005159858581290555), np.float64(6.272576290401881e-14),
    ...np.float64(2.8320201318030544e-05)), (np.float64(2.296393963305256e-19),
    ...np.float64(0.5000000000002758), np.float64(0.2777784238527222),
    ...np.float64(0.22222157614700191)))
shares: Einstein > 0.99 at Ha = 1e-2, Gauss-Bonnet > 0.9 at Ha = 30, Gauss-Bonnet and cubic
    ...about 1/2 each at Ha = 3e5 (to 1e-2), gas > 0.1 at Ha = 1e15 : True
coupled: where the gas is a small fraction, the effective exponents are those of the vacuum
    ...solution (1, p2, p1 within 1e-2) : True
NUM coupled, gas-dominated late regime (Ha = 1e-14): tau Ha, tau Hb, pa/eps, pb/eps :
    ...(np.float64(0.00031495953861375523), np.float64(0.3999952443921748),
    ...np.float64(1.873595711619992e-06), np.float64(0.24999859480321623))
without a stabilizing potential the gas-dominated Einstein regime decompactifies: a -> const
    ...(tau Ha -> 0), b ~ tau^(2/5) (tau Hb -> 2/5), pb/eps -> 1/4 : True
NUM gas-fraction growth exponent in the Einstein regime (numerical, between Ha = 1e-2 and
    ...1e-4), and 2 - 3p1 - 5q1 : (1.1698331157808215, 1.1601886205085203)
gas fraction grows like tau^(2 - 3p1 - 5q1) = tau^1.160 in the Einstein regime (to 2e-2) : True
Einstein + fluid power laws a ~ tau^P, b ~ tau^Q: (wa, wb) = (0, 1/4) -> (P, Q) = (0, 2/5)
    ...[decompactification]; (1/3, 0) -> (1/2, 0) [brane radiation, b frozen]; (1/7, 1/7) ->
    ...(1/4, 1/4) [isotropic] : True
matter-dominated cubic regime (any fluid pa = w_a eps, b finite): leading 44 equation p^3 (p -
    ...1)(648 alpha3 h^2 - 12 alpha2) = 0 -> h^2 = alpha2/(54 alpha3); 11/00 -> p = 1/(1 +
    ...w_a); kappa eps tau^3 = (64/3) p^3 alpha2 |h| : True
NUM h* = -sqrt(alpha2/(54 alpha3)), h0 = -sqrt(alpha2/(30 alpha3)) (xi = 1e-6) :
    ...(-136.08276348795434, -182.57418583505537)
NUM Einstein-Kasner pair creation |beta|^2 = F(K) at K = 0.3, 1, 3, 8, 16, and the tail
    ...exponent d ln F/d ln K between 8 and 16 : ((0.13558706527933204, 0.010749346167248614,
    ...0.0001493265403051232, 7.104066464566534e-06, 8.157299133414563e-07),
    ...-3.1224816166636584)
pair creation has a power-law tail F ~ K^(-2(p1 - q1)/(1 - p1)) = K^(-(3 sqrt 2 + 2)/2) =
    ...K^(-3.121) (to 2e-2): it comes from the singularity, so the number of created pairs is
    ...dominated by the earliest times : True
F(1) agrees with the value of section 14.3 of 7+1.nb (|beta|^2 for (|p|, |q|) = (1, 1)) : True
```

```
NUM brane radiation fraction kappa eps_W/(3 H^2) at T = M8, H = M8 (M8 = 10 TeV, reduced M_Pl =
    ...2.435e18 GeV, g* = 106.75) : 1.9743638405361938e-28
the Standard Model plasma on the brane (T <= M8) is gravitationally negligible in the bulk
    ...while H ~ M8: fraction < 1e-27 : True
NUM capture: start tau, beta(start) = ln(b/b0), m_r : (np.float64(56.750492072167184),
    ...np.float64(0.4595957216796004), 0.001)
NUM after capture: tau_E from <H_E> over the period after tau = 2.9e5, then beta = A sin(m_r
    ...tau + vartheta)/(m_r (tau - tau_E)) fitted on 1e5 <= tau <= 3e5: A, vartheta, tau_E, max
    ...residual/amplitude : (0.46933049563194473, 2.136009270851999,
    ...np.float64(-1564.3580565850134), np.float64(0.012786248665086845))
NUM after capture: <H_E>(tau - tau_E) over one period at tau = 1e5, 2e5 (H_E = Ha + 2Hb,
    ...four-dimensional matter domination: 2/3) : (np.float64(0.6669420716555031),
    ...np.float64(0.6667362091499506))
NUM after capture: gas fraction kappa eps/(3 H_E^2) at tau = 1e5 and 3e5 (constant: both are
    ...four-dimensional matter) : (np.float64(0.008595792799709226),
    ...np.float64(0.008728032079226138))
capture: the radion oscillates as a four-dimensional massive field in a matter-dominated
    ...universe: j0 form to 2e-2, <H_E>(tau - tau_E) -> 2/3 to 1e-2, the gas fraction stays
    ...constant to 2e-2 : True
NUM after capture: radion share of the energy, 1 - gas fraction (the cosmological moduli
    ...problem) : 0.9912719679207739
four-dimensional (Einstein-frame) form of the 00 equation: 3 (Ha + 2Hb)^2 = 6 Hb^2 + kappa
    ...eps_total, i.e. 3 H_E^2 M_Pl^2 = (1/2) phi'^2 + rho with phi = sqrt(12) M_Pl ln(b/b0) :
    ...True
NUM radion of mass 10 GeV (M8 = 10 TeV): decay rate into brane fields (b, c, tau, mu pairs and
    ...two gluons) in GeV, lifetime in s, reheating temperature in GeV :
    ...(4.572168576618335e-37, 1439605617925.0205, 1.012614577110346e-09)
through brane fields alone a radion of 10 GeV would decay far too late (T_RH < 1 keV < 4 MeV
    ...needed by nucleosynthesis): the moduli problem of the minimal model : True
NUM number of Kaluza-Klein momenta of the auxiliary torus with |q| < m_r/2 (m_r = 10 GeV):
    ...V_perp (pi^2/2)(m_r/2)^4/(2 pi)^4 : 11733517482065.24
Jacobi's four-square formula r_4(k) = 8 sigma(k) - 32 sigma(k/4) equals the direct count of
    ...lattice vectors for 1 <= k <= 60 : True
NUM radion into bulk dirac16complex pairs (m << m_r/2): Gamma_bulk lattice, continuum (GeV),
    ...lifetime (s), Gamma_bulk/Gamma_brane : (5.9991693514127285e-24,
    ...np.float64(5.9991693514135954e-24), 0.10971718222040179, 13121058969898.768)
the radion decays into the bulk: lattice sum = continuum limit to 1e-6, number of momenta =
    ...N_KK to 1e-5, Gamma_bulk/Gamma_brane > 1e12 : True
NUM M8, M_Pl (reduced), V_perp = M_Pl^2/M8^6 (GeV^-4), l_perp = V_perp^(1/4) (m), lowest
    ...Kaluza-Klein mass 2 pi/l_perp (GeV) : (10000.0, 2.435e+18, 5929225000000.0,
    ...3.079186208511254e-13, 0.004026524867292262)
NUM g*(T), g*s(T) at T = 1 TeV, 10 GeV, 1 GeV, 300 MeV, 100 MeV, 10 MeV, 1 MeV, 100 keV, 1 keV
    ...: ((106.79411834061365, 106.77847184722867), (86.29628742994949, 86.20451456036336),
    ...(76.41793064429221, 75.57705711354103), (64.64582342667666, 64.2172398064234),
    ...(18.315240842717653, 17.94958998859558), (10.837168710423134, 10.835178439973458),
    ...(10.633559462970378, 10.636104471804657), (4.327020814133464, 4.814383599639942),
    ...(3.383002311703709, 3.9374638708790815))
g* limits: 106.75 above the electroweak scale (to 0.1), 10.75 near 3 MeV (to 0.1), and after
    ...e+e- annihilation 2 + (21/4)(4/11)^(4/3)(N_eff/3) = 3.3855 (to 1e-3) : True
NUM radiation era: t(T) in s (entropy conservation with g*(T), g*s(T)) at T = 1 GeV, 100 MeV,
    ...10 MeV, 1 MeV, 100 keV, 10 keV : (np.float64(2.794095659821038e-07),
    ...np.float64(6.123536906169378e-05), np.float64(0.007393002790464219),
    ...np.float64(0.7431868178969353), np.float64(118.83951905287026),
    ...np.float64(13161.662106152166))
NUM radiation era: ratio of the instantaneous approximation t = 1/(2H(T)) to t(T) :
    ...(np.float64(0.9907571426533383), np.float64(0.9234161431082585),
    ...np.float64(0.9943226486914915), np.float64(0.9985476394981425),
    ...np.float64(0.978927979398272), np.float64(0.9996413686006429))
NUM radiation era: relative change of t(10 keV) when the grid is halved (error estimate) :
    ...3.579397018782515e-07
radiation era: the t(T) quadrature is converged (halving the grid changes t(10 keV) by < 1e-5)
    ...: True
radiation era: t(1 MeV) = 0.74 s within 5% (standard value 1/(2H) with g* = 10.75) : True
NUM late universe: Omega_r, Omega_Lambda, a_eq = Omega_r/Omega_m, z_eq, age (Gyr) :
    ...(9.209579083825086e-05, 0.6849079042091618, 0.00029236758996270117, 3419.351756935764,
```

```
    ...13.790686840671615)
age of the universe 13.79 Gyr (Planck 2018: 13.787 +- 0.020) : True
NUM composite solution t(a) = t_rm(a) + t_mL(a) - t_m(a): maximal relative error for a =
    ...10^(-8), 10^(-7.75), ..., 10^0.5 : 0.000220157276312416
the composite analytic solution (exact radiation+matter, exact matter+Lambda, minus their
    ...common matter part) is accurate to < 3e-4 at all times : True
NUM radion displacement today (m_r = 10 GeV): from Lambda4 + matter, from a brane tension f_W =
    ...M8^4 : (1.741296272562016e-86, 5.6218701994499e-24)
NUM Gauss-Bonnet and cubic corrections today: alpha2 H0^2 and alpha3 H0^4 with alpha2 = 1/M8^2,
    ...alpha3 = 1e-6/M8^4 : (2.0670464555964908e-92, 4.2726810495940146e-190)
NUM normalcy temperature (order of magnitude) T* = (M8^6/M_Pl)^(1/5) in GeV : 13.264784746562139
NUM far future: H_inf = H0 sqrt(Omega_Lambda) (1/s and GeV), Gibbons-Hawking temperature H/(2
    ...pi) in K : (1.80769618063081e-18, 1.1898472405336613e-42, 2.1975484073507053e-30)
far future: H(t) = H_inf coth(3 H_inf t/2) (exact for matter + Lambda) agrees with the full
    ...solution at a = 3, 10 to 1e-4 : True
far future: the radion obeys beta'' + gamma beta' + m_eff^2 beta = 0 around de Sitter
    ...(constraint used for delta Ha); alpha2 = alpha3 = 0: gamma = 3H, m_eff^2 = m_r^2 at
    ...beta_eq -> 0 : True
NUM far future (beta_eq -> 0): gamma/(3H) and m_eff^2/m_r^2 with Gauss-Bonnet and cubic terms
    ...(sympy) : ('-(1024*H**8*alpha2**4 - 512*H**6*alpha2**3 - 192*H**6*alpha2*alpha3 -
    ...48*H**4*alpha2**2 + 48*H**4*alpha3 + 16*H**2*alpha2 - 1)/(32*H**4*alpha2**2 -
    ...4*H**2*alpha2 + 1)**2', '1/(32*H**4*alpha2**2 - 4*H**2*alpha2 + 1)')
far future: equilibrium of the radion in de Sitter: 3 m_r^2 beta_eq = 3H^2 - 12 alpha2 H^4 (+
    ...O(beta_eq^2)) : True
NUM far future: 2 pi M/H_inf for M = 1 eV and 1 GeV (|beta|^2 = 1/(exp(2 pi M/H) + 1)) :
    ...(5.280665528426573e+33, 5.280665528426573e+42)
eight-dimensional de Sitter (no stabilizer): 21 H^2 - 420 alpha2 H^4 + 2520 alpha3 H^6 = Lambda
    ...(all three equations) : True
NUM de Sitter radion test (l2 = 1, Lambda = 3e-4, m_r = 0.1, start 0.05 above): beta(3000),
    ...Ha(3000), and the equilibrium (beta_eq, H_eq) of the two algebraic equations :
    ...(np.float64(0.010204078087430525), np.float64(0.010103586699399756),
    ...0.010204078087430532, 0.010103586699399753)
far future: the radion relaxes to its shifted equilibrium (to 1e-6 relative) and Ha to H_eq (to
    ...1e-8) : True
boolean checks: 50   failed: 0
VERDICT: ALL CHECKS PASSED
```